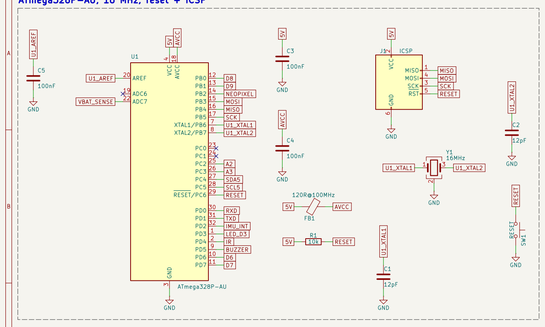
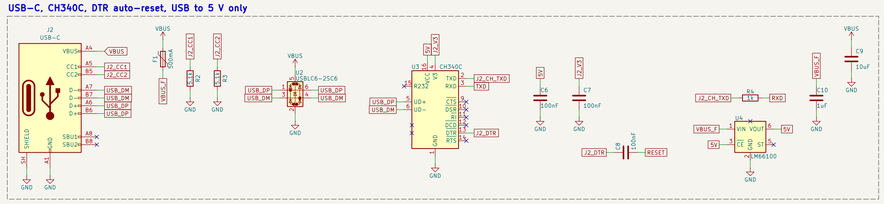
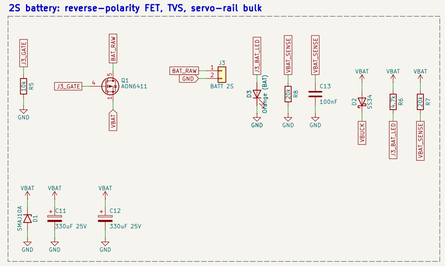
ATmega328P-AU, 16 MHz, reset + ICSP



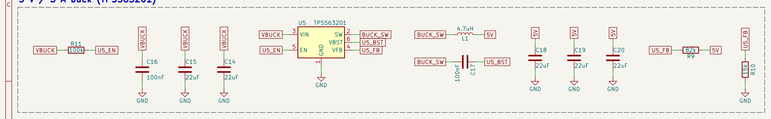
USB-C, CH340C, DTR auto-reset, USB to 5 V only



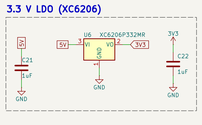
2S battery: reverse-polarity FET, TVS, servo-rail bulk



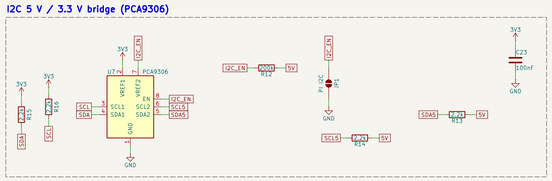
5 V / 3 A buck (TPS563201)



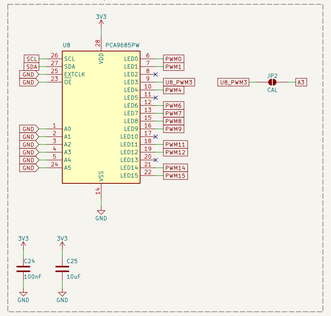
3.3 V LDO (XC6206)



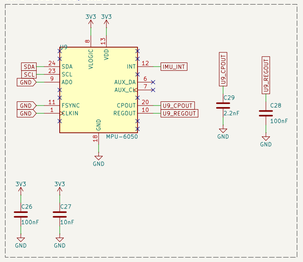
I2C 5 V / 3.3 V bridge (PCA9306)



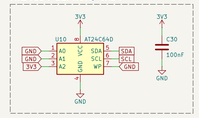
PCA9685 servo PWM, 0x40



MPU-6050 IMU, 0x68



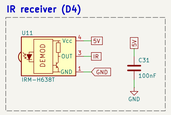
AT24C64 EEPROM, 0x54



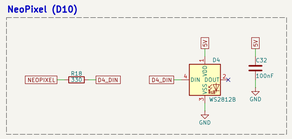
Piezo buzzer (D5)



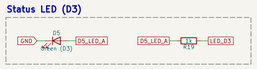
IR receiver (D4)



NeoPixel (D10)



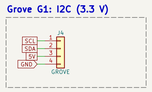
Status LED (D3)



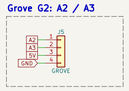
5 V power LED



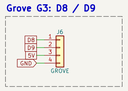
Grove G1: I2C (3.3 V)



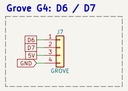
Grove G2: A2 / A3



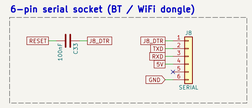
Grove G3: D8 / D9



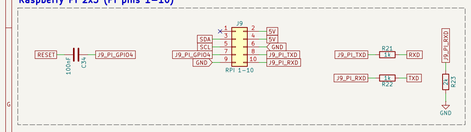
Grove G4: D6 / D7



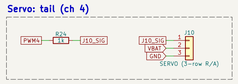
6-pin serial socket (BT / WiFi dongle)



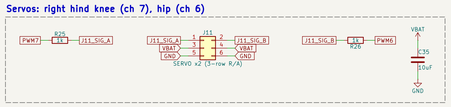
Raspberry Pi 2x5 (Pi pins 1–10)



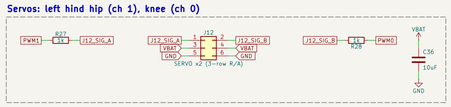
Servo: tail (ch 4)



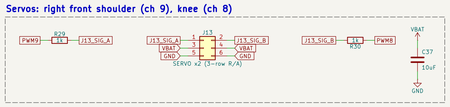
Servos: right hind knee (ch 7), hip (ch 6)



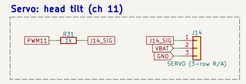
Servos: left hind hip (ch 1), knee (ch 0)



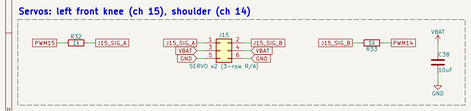
Servos: right front shoulder (ch 9), knee (ch 8)



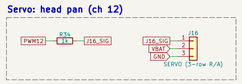
Servo: head tilt (ch 11)



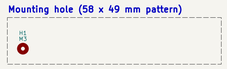
Servos: left front knee (ch 15), shoulder (ch 14)



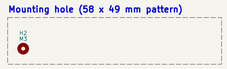
Servo: head pan (ch 12)



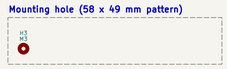
Mounting hole (58 x 49 mm pattern)



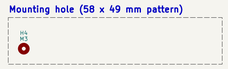
Mounting hole (58 x 49 mm pattern)



Mounting hole (58 x 49 mm pattern)



Mounting hole (58 x 49 mm pattern)



Power flags



Generated from blocks → design.json by heypcb-kicad

HeyPCB method / pure KiCad

Sheet: /

File: whisker.kicad_sch

Title: Whisker — controller for the Petoi Nybble robot cat, compatible with th

Size: A1	Date: 2026-10-02
KiCad E.D.A. 10.0.6	**Rev: A**
	Id: 1/1